Exercises

A14-N01

Compute gcd(252,198), listing each quotient and remainder.

A14-N02

Use extended Euclid to find Bézout coefficients for 252 and 198.

A14-N03

Find all solutions to $14x \equiv 30 \pmod{100}$.

A14-N04

Find all solutions to $6x \equiv 5 \pmod{9}$.

A14-N05

Solve $x \equiv 2 \pmod{6}$, $x \equiv 5 \pmod{9}$.

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A14-N06

Decide whether $x \equiv 2 \pmod{6}$, $x \equiv 4 \pmod{9}$ is solvable.

A14-N07

Compute $7^{13} \bmod 23$ bit by bit and explain $e=0$.

A14-N08

Run one Miller–Rabin round on $n=15$ with $a=2$.

A14-N09

For $p=11$, $q=13$, $e=7$ find the RSA private exponent.

A14-N10

Encrypt $M=9$ with that key and decrypt it.

A14-X01

List Euclid’ s remainders for gcd(55,34), explain the connection between quotient 1 and Fibonacci numbers, and distinguish division count from bit cost.

A14-X02

Construct and check the CRT solution of $x\equiv 1 \pmod 4$, $x\equiv 2 \pmod 5$, $x\equiv 3 \pmod 7$.

A14-X03

Solve $x\equiv 1 \pmod 4$, $x\equiv 3 \pmod 6$, then decide whether $x\equiv 1 \pmod 4$, $x\equiv 2 \pmod 6$ is solvable.

A14-X04

Find $\mathrm{ord}_{11}(2)$, decide whether 2 is a primitive root, and solve $2^x\equiv 7 \pmod{11}$.

A14-X05

For n=9 list every Miller–Rabin strong liar in $1\le a\le 8$, and compare the 1/4 bound.

A14-X06

For n=91, $f(x)=x^2+1 \bmod 91$, and slow=fast=2, run one Floyd round and compute the gcd. Explain what to do if the gcd equals n.

A14-X07

For n=8051, $x_0=2$, and $f(x)=x^2+1 \bmod n$, list the first three Floyd rounds and the checkpoint comparisons with $y=x_1$, x_2, x_4, and identify when each obtains a factor. Give the no-collision upper bound for k independent uniform draws from p=97 values, and explain its significance for fixed f.

A14-X08

List U(12) and the square of each element and decide whether it is cyclic. Find a primitive root modulo 9 and list its powers.

A14-X09

Verify that x=4 is a nontrivial square root of 1 modulo 15, then obtain factors using the two gcds.

A14-X10

Let p=5, q=11, e=3 and d=27. Encrypt the non-unit message m=5, decrypt using a square chain, and explain the result in CRT coordinates.

A14-X11

For the previous ciphertext c, choose r=2, compute $c' = c \cdot r^e \bmod n$ and decrypt it. What property of textbook RSA does this illustrate?

A14-X12

In the additive group Z_{12} let H={0,4,8}. List all distinct cosets and verify Lagrange's theorem. Then compute $\phi(72)$ using prime powers and multiplicativity.

A14-X13

Find all square roots of 1 modulo 25 and explain why the same two-root conclusion cannot be directly applied modulo 15.

A14-X14

For textbook RSA signatures with public key n=55, e=3, choose σ=12 and construct a message m that verifies. Does this let the attacker sign an arbitrary prescribed message?

A14-X15

Verify the factorization of 341 and its base-2 Fermat result. Then give d, s, and the Miller–Rabin squaring sequence and decide whether it passes.

A14-X16

Explain why every base coprime to 561 passes its Fermat test. Then compute its base-2 Miller–Rabin sequence and decide.

A14-X17

Verify that 2047 is composite, and use the congruence for 2^{11} to explain why its base-2 Miller–Rabin round passes. Does a single fixed-base-2 test have the randomized 1/4 error guarantee?

A14-X18

State the right-to-left modular exponentiation invariant and trace $7^{19} \bmod 23$. Explain exponent 0, modulus 1, and non-coprime bases.

A14-X19

Solve $24x \equiv 36 \pmod{60}$ and encode all solutions as a tuple. Also encode $0x \equiv 0 \pmod 9$, distinguishing compact representation from explicit output cost.

A14-X20

Prove that 2 is a primitive root modulo 121; listing its order modulo 11 is insufficient.

A14-X21

For n=65 list all strong liars layer by layer using CRT, calculate the passing fraction when bases are sampled from 2,…,63, and classify base 2.

A14-X22

For independently and uniformly sampled Miller–Rabin bases, find the smallest number of rounds certified by the universal bound 4^{-k} to make each fixed composite pass with probability at most 10^{-9}. If the prior prime probability is $\pi=1/100$ and primes always pass, bound the posterior probability of a composite given a pass. Explain repeated fixed-base tests.

A14-X23

Prove Floyd’s meeting bound in terms of μ and λ on a finite function graph. For n=15, $x_0=0$, $f(x)=x^2+2$ compute each gcd until a collision and explain retry and inconclusive outcomes.

A14-X24

Prove Euclid halves the second argument within two divisions, and give a Bézout certificate for (391,299). Does 299 have an inverse modulo 391?

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd ed., Chapter 31. MIT 6.046J Fall 2004 Lecture 7, p. 33, and Fall 2005 Lecture 8, p. 13: inverses modulo a prime.

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